20 Outline two ethical positions relating to whether the use of performance-enhancing substances by athletes is acceptable.

4.4 Topic 8: Grey matter

Students will be assessed on their ability to:

1. Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 13 of this specification.
2. Describe how plants detect light using photoreceptors and how they respond to environmental cues.
3. Describe the structure and function of sensory, relay and motor neurones including the role of Schwann cells and myelination.
4. Describe how a nerve impulse (action potential) is conducted along an axon including changes in membrane permeability to sodium and potassium ions and the role of the nodes of Ranvier.
5. Describe the structure and function of synapses, including the role of neurotransmitters, such as acetylcholine.
6. Describe how the nervous systems of organisms can detect stimuli with reference to rods in the retina of mammals, the roles of rhodopsin, opsin, retinal, sodium ions, cation channels and hyperpolarisation of rod cells in forming action potentials in the optic neurones.
7. Explain how the nervous systems of organisms can cause effectors to respond as exemplified by pupil dilation and contraction.
8. Compare mechanisms of coordination in plants and animals, ie nervous and hormonal, including the role of IAA in phototropism (details of individual mammalian hormones are not required).
9. Locate and state the functions of the regions of the human brain's cerebral hemispheres (ability to see, think, learn and feel emotions), hypothalamus (thermoregulate), cerebellum (coordinate movement) and medulla oblongata (control the heartbeat).

10 Describe the use of magnetic resonance imaging (MRI), functional magnetic resonance imaging (fMRI) and computed tomography (CT) scans in medical diagnosis and investigating brain structure and function.

11 Discuss whether there exists a critical 'window' within which humans must be exposed to particular stimuli if they are to develop their visual capacities to the full.

12 Describe the role animal models have played in developing explanations of human brain development and function, including Hubel and Wiesel's experiments with monkeys and kittens.

13 Consider the methods used to compare the contributions of nature and nurture to brain development, including evidence from the abilities of newborn babies, animal experiments, studies of individuals with damaged brain areas, twin studies and cross-cultural studies.

14 Describe how animals, including humans, can learn by habituation.

15 **Describe how to investigate habituation to a stimulus**.

16 Discuss the moral and ethical issues relating to the use of animals in medical research from two ethical standpoints.

17 Explain how imbalances in certain, naturally occurring, brain chemicals can contribute to ill health (eg dopamine in Parkinson's disease and serotonin in depression) and to the development of new drugs.

18 Explain the effects of drugs on synaptic transmissions, including the use of L-Dopa in the treatment of Parkinson's disease and the action of MDMA in ecstasy.

19 Discuss how the outcomes of the Human Genome Project are being used in the development of new drugs and the social, moral and ethical issues this raises.

20 Describe how drugs can be produced using genetically modified organisms (plants and animals and micro organisms).

21 Discuss the risks and benefits associated with the use of genetically modified organisms.

CONTEXT-LED APPROACH BASED ON THE SALTERS-NUFFIELD ADVANCED BIOLOGY PROJECT

The following section shows how the specification may be taught using a context-led approach. The outcomes in this section are presented in a different order to those in the section that contains the concept approach and therefore the outcomes do not appear in numerical order.